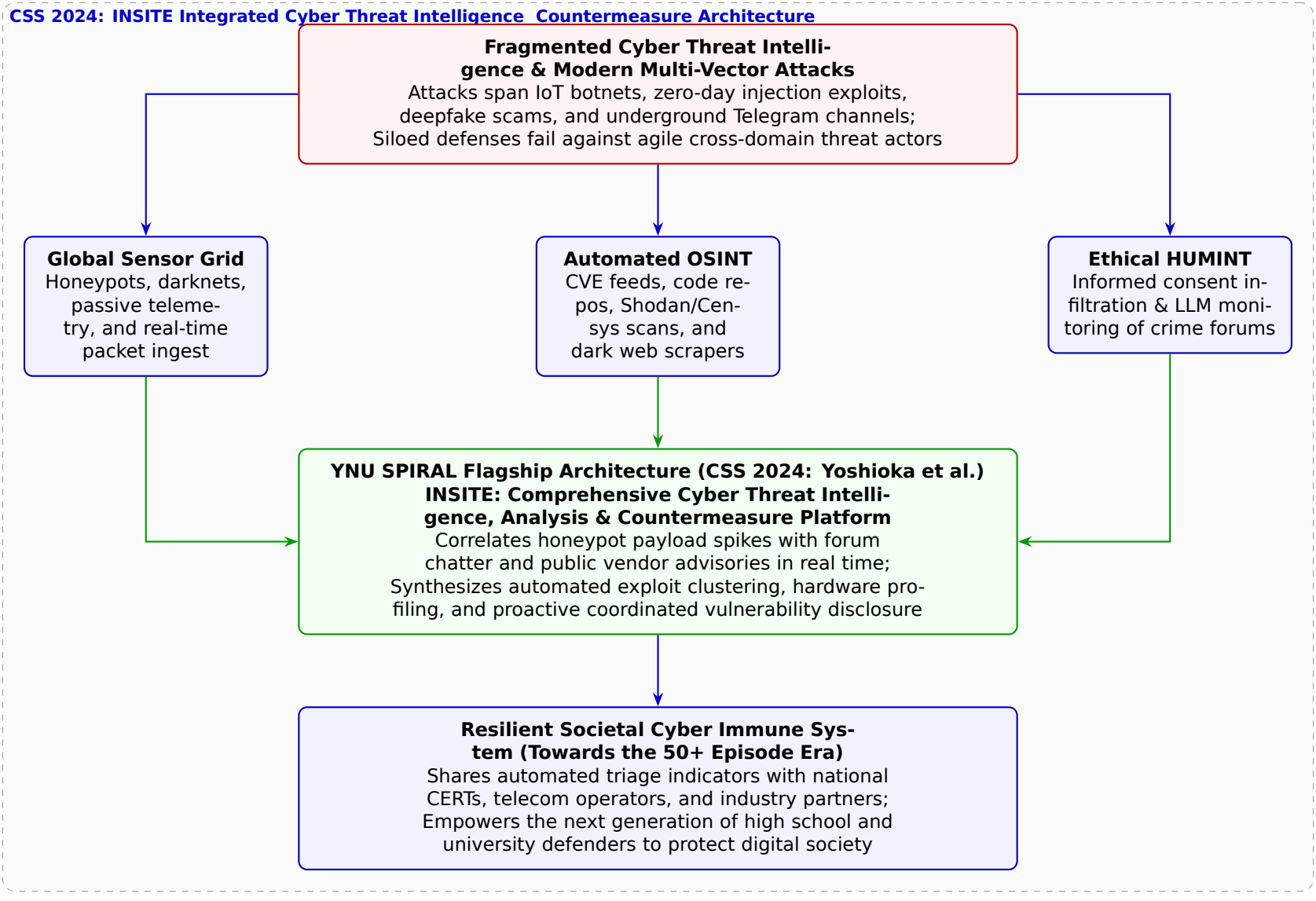
CSS 2024: INSITE Integrated Cyber Threat Intelligence Countermeasure Architecture
Fragmented Cyber Threat Intelligence & Modern Multi-Vector Attacks
Attacks span IoT botnets, zero-day injection exploits, deepfake scams, and underground Telegram channels; Siloed defenses fail against agile cross-domain threat actors
Global Sensor Grid
Honeypots, darknets, passive telemetry, and real-time packet ingest
Automated OSINT
CVE feeds, code repos, Shodan/Censys scans, and dark web scrapers
Ethical HUMINT
Informed consent infiltration & LLM monitoring of crime forums
YNU SPIRAL Flagship Architecture (CSS 2024: Yoshioka et al.)
INSITE: Comprehensive Cyber Threat Intelligence, Analysis & Countermeasure Platform
Correlates honeypot payload spikes with forum chatter and public vendor advisories in real time; Synthesizes automated exploit clustering, hardware profiling, and proactive coordinated vulnerability disclosure
Resilient Societal Cyber Immune System (Towards the 50+ Episode Era)
Shares automated triage indicators with national CERTs, telecom operators, and industry partners; Empowers the next generation of high school and university defenders to protect digital society